Sentences that say several things at once

A short fictional research summary describing the TCGA-BRCA files (release A). Each sentence states two or more things; the first group is true, the second has one wrong part each, the third cannot be read without guessing.

True

CDH1 was mutated in 59% of lobular carcinomas but only 1.6% of ductal carcinomas, and PIK3CA mutations occurred in 48.5% of Luminal A tumors versus 6.6% of Basal-like tumors. Among HER2-positive tumors, about 54% carry an ERBB2 amplification, compared with under 2% of HER2-negative tumors. The cohort includes 12 male patients, all of them ER-positive. The 1,101 primary tumors came from 1,085 women and 12 men, with sex missing for 4. Death was most frequent in stage IV disease, where 15 of 20 patients died. PAM50 RNA-seq calls are available for 844 of the 1,097

primary tumors with RNA data. Mean HER2 protein was highest in the HER2-enriched subtype. HER2 immunohistochemistry was equivocal for 179 primary tumors. GATA3 mutations were found only in ER-positive tumors. Lobular carcinomas were 18.5% of tumors, and CDH1 was mutated in 59% of them.

One wrong part

TP53 was mutated in 80% of Basal-like tumors but only 10.3% of Luminal A tumors. Among ER-positive tumors, about 30% carry a PIK3CA mutation, compared with 15% of ER-negative tumors. The study has 12 male patients, all of them ER-negative. Overall, 1,101 primary tumors came from 1,085 women and 12 men, with sex missing for 9. Death was most frequent in stage I disease, where 16 of 183 patients died. Mean HER2 protein was highest in the Luminal A subtype. HER2 immunohistochemistry was equivocal for 250 primary tumors. TP53 mutations were found only in ER-negative tumors. Ductal carcinomas were

71.5% of tumors, and PIK3CA was mutated in 60% of them.

Cannot be read without guessing

CDH1 and PIK3CA were mutated in 59% of lobular carcinomas and 48.5% of Luminal A tumors. The cohort has 12 male patients, 50% of them ER-negative. These differences likely reflect screening practice. TP53 mutations cluster in the DNA-binding domain.